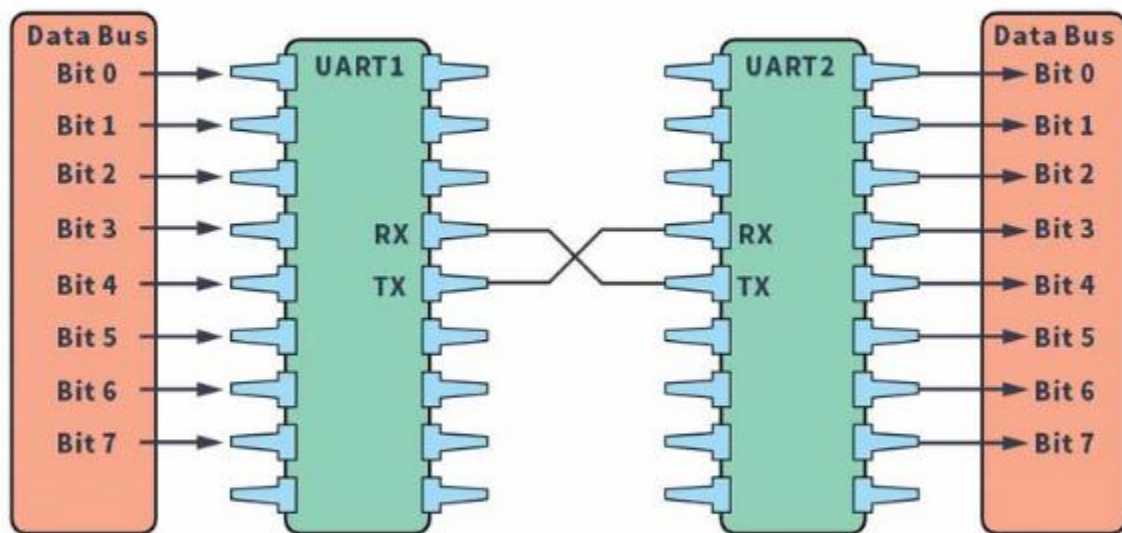


UART_Project

UART Communication



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Overview:

1. RTL DATA FLOW & TOP-LEVEL ARCHITECTURE

The UART Transmitter (UART_TX) is a parameterizable digital IP designed to convert parallel input data (P_INPUT) into an asynchronous serial bitstream (TX_OUTPUT).

The hardware datapath and control flow proceed as follows:

A. IDLE State & Parallel Load:

- When the system is IDLE and a valid data flag (V_INPUT = 1) is asserted, the Main_controller_FSM asserts the internal 'load' signal.
- The Serializer module samples 'P_INPUT' into its internal shift register (shift_reg <= P_INPUT) and resets its internal bit counter to zero.

B. Transmission Framing & State Sequence:

- **START State:** The FSM drives 'mux_sel = 2'b00', routing 'start_bit' (1'b0) to TX_OUTPUT via the output multiplexer.
- **DATA State:** The FSM asserts 'ser_en = 1' and 'mux_sel = 2'b01'.
 - The Serializer outputs shift_reg[0] (LSB first) onto 'serial_bit'.
 - On each clock cycle, shift_reg shifts right by 1 bit, and counter increments until all N bits are shifted out.
 - Upon shifting the last bit, Serializer asserts the 'ser_done' pulse.
- **PARITY State (Optional):** If parity is enabled (P_EN = 1), the FSM transitions to PARITY ('mux_sel = 2'b10').
 - The paritybit_calc module calculates even parity (XOR) if P_BIT = 0 or odd parity (XNOR) if P_BIT = 1 over P_INPUT, driving 'parity_bit' to TX_OUTPUT.
- **STOP State:** The FSM drives 'mux_sel = 2'b11', routing 'stop_bit' (1'b1) to TX_OUTPUT.
 - **Handshaking:** If V_INPUT is asserted high during the STOP bit, 'load' is asserted immediately to pre-load the next data payload for back-to-back framing.

2. SUBMODULE INTERCONNECTIVITY:

- **Main_controller_FSM:** Controls state progression and generates control signals (load, ser_en, mux_sel, BUSY) based on inputs V_INPUT, P_EN, and ser_done.

- **Serializer:** Parameterized payload shifter generating 'serial_bit' and 'ser_done'.
- **paritybit_calc:** Combinational parity generator outputting 'parity_bit'.
- **mux:** Output routing multiplexer driving final 'TX_OUTPUT'.

3. VERIFICATION & TEST CASE STRATEGY:

To fully verify the UART_TX module, testbenches typically evaluate the following scenarios:

A. Reset Verification:

- Assert reset ($RST = 0$) to ensure FSM returns to IDLE, BUSY is deasserted (0), and shift counters are zeroed.

B. Parity Mode Verification (Even & Odd):

- Even Parity ($P_EN = 1, P_BIT = 0$): Verify parity bit equals XOR of P_INPUT .
- Odd Parity ($P_EN = 1, P_BIT = 1$): Verify parity bit equals XNOR of P_INPUT .

C. Standard Frame Transmissions (Parity Disabled):

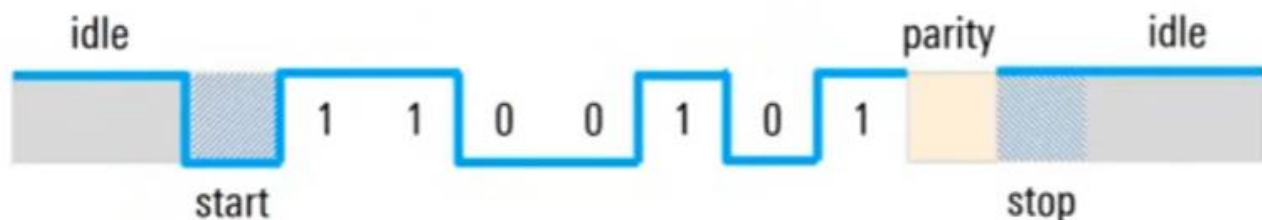
- Set $P_EN = 0$, apply a valid payload (P_INPUT) with $V_INPUT = 1$.
- Verify start bit (0), N data bits (LSB to MSB), and stop bit (1) timing.

D. Back-to-Back Transmissions:

- Assert V_INPUT while the current frame is in the STOP state.
- Verify that 'load' asserts immediately, bypassing IDLE to transmit the next frame without line delay.

E. IDLE Line Stability:

- Ensure TX_OUTPUT holds high (1'b1) continuously while IDLE and BUSY remains 0.



RTL_Code:

Main_Controller_FSM:

```
module Main_controller_FSM #(
    //input size
    parameter N = 8
)(
    //inputs
    input V_INPUT,
    input P_EN,
    input CLK,RST,
    input ser_done,
    //outputs
    output reg load,
    output reg ser_en,
    output reg [1:0]mux_sel,
    output reg BUSY
);
//states
parameter IDLE = 3'b000;
parameter START = 3'b001;
parameter DATA = 3'b010;
parameter PARITY = 3'b011;
parameter STOP = 3'b100;
//state signals
reg [2:0] cs,ns;

//state memory
always@(posedge CLK or negedge RST)begin
    if(!RST)begin
        cs <= IDLE;
    end
    else begin
        cs <= ns;
    end
end

//next_state_logic
always@(*)begin
    case(cs)
        3'b000:begin //IDLE
            if(V_INPUT == 1)begin
                ns = START;
            end
        end
    endcase
end
```

```

        end
        else begin
            ns = IDLE;
        end
    end
end
3'b001:begin //START
    ns = DATA;
end
3'b010:begin //DATA
    if(ser_done)begin
        if(P_EN)begin
            ns = PARITY;
        end
        else begin
            ns = STOP;
        end
    end
    else begin
        ns = DATA;
    end
end
3'b011:begin //PARITY
    ns = STOP;
end
3'b100:begin //STOP
    if(V_INPUT == 1)begin
        ns = START;
    end
    else begin
        ns = IDLE;
    end
end
default : ns = IDLE;
endcase
end
//output logic
always@(*)begin
    BUSY =1;
    load=0;
    ser_en=0;
    mux_sel=2'b11;
    case(cs)
        3'b000: begin //IDLE
            BUSY = 0;
            mux_sel =2'b11;
            if(V_INPUT)

```

```
        load = 1;
    end
    3'b001:begin //START
        mux_sel =2'b00;
    end
    3'b010:begin //DATA
        ser_en=1;
        mux_sel =2'b01;
    end
    3'b011:begin //PARITY
        mux_sel =2'b10;
    end
    3'b100:begin //STOP
        mux_sel =2'b11;
        if(V_INPUT)
            load = 1;
        end
    default: begin
        BUSY =0;
        load=0;
        ser_en=0;
        mux_sel=2'b11;
    end
endcase
end
endmodule
```

Serializer:

```
module Serializer#(
    parameter N = 8
)(
    input [N-1:0] P_INPUT,
    input CLK,RST,
    input ser_en,load,
    output serial_data_o,
    output reg ser_done
);
reg [N-1:0]shift_reg;
reg [$clog2(N):0]counter;
//
assign serial_data_o = shift_reg[0]; //transmit LSB first
//
always@(posedge CLK or negedge RST)begin
    if(!RST)begin
        shift_reg <= 0;
        counter <= 0;
        ser_done<=0;
    end
    else begin
        if(load)begin
            shift_reg <= P_INPUT;
            counter <= 0;
            ser_done <= 0;
        end
        else if(ser_en)begin
            if(counter < N-1)begin
                shift_reg <= shift_reg >> 1;
                counter <= counter + 1;
                ser_done<=0;
            end
            else if(counter == N-1) begin
                shift_reg <= shift_reg >> 1;
                counter <= counter + 1;
                ser_done <=1;
            end
            else begin
                ser_done <= 0;
            end
        end
        else begin
            counter <= 0;
            ser_done <= 0;
        end
    end
end
```

```

        end
    end
end
endmodule

```

Parity_bit_calculator:

```

module paritybit_calc#(
    parameter N = 8
)(
    input P_BIT,
    input [N-1:0] P_INPUT,
    output reg parity_bit
);
always @(*) begin
    case (P_BIT)
        1'b0: parity_bit = ^ P_INPUT; //even
        1'b1: parity_bit = ~^ P_INPUT; //odd
        default : parity_bit = 0;
    endcase
end
endmodule

```

MUX_4x1:

```

module mux(
    input start_bit,
    input serial_bit,
    input parity_bit,
    input stop_bit,
    input [1:0]mux_sel,
    output TX_OUTPUT
);
assign TX_OUTPUT = (mux_sel==2'b00)? start_bit:
                   (mux_sel==2'b01)? serial_bit:
                   (mux_sel==2'b10)? parity_bit: stop_bit;
endmodule

```

TOP_Module:

```
module UART_TX #(
    parameter N = 8
)(
    input [N-1:0]P_INPUT,
    input V_INPUT,
    input CLK,
    input RST,
    input P_EN,
    input P_BIT,
    output TX_OUTPUT,
    output BUSY
);

    wire ser_done;
    wire load;
    wire ser_en;
    wire [1:0] mux_sel;
    wire serial_bit;
    wire parity_bit;
    wire start_bit;
    wire stop_bit;
    assign start_bit = 1'b0;
    assign stop_bit = 1'b1;
    //////////////////////////////////////////instantiation////////////////////////////////////////

    Main_controller_FSM #(.N(N))
fsm(.V_INPUT(V_INPUT),.P_EN(P_EN),.CLK(CLK),.RST(RST),.ser_done(ser_done),
    .BUSY(BUSY),.load(load),.ser_en(ser_en),.mux_sel(mux_sel));

    Serializer #(.N(N)) ser(.P_INPUT(P_INPUT),.CLK(CLK),.RST(RST),.ser_en(ser_en),.load(load),
    .serial_data_o(serial_bit),.ser_done(ser_done));

    paritybit_calc #(.N(N)) pc(.P_INPUT(P_INPUT),.P_BIT(P_BIT),.parity_bit(parity_bit));

    mux
mux_inst(.start_bit(start_bit),.serial_bit(serial_bit),.parity_bit(parity_bit),.stop_bit(stop_bit),.
mux_sel(mux_sel),.TX_OUTPUT(TX_OUTPUT));
endmodule
```

Testbench:

```
module UART_TX_tb();
    parameter N = 8;
    reg [N-1:0]P_INPUT;
    reg V_INPUT;
    reg CLK;
    reg RST;
    reg P_EN;
    reg P_BIT;
    wire TX_OUTPUT;
    wire BUSY;
    //
    UART_TX #(N(N))
dut(.P_INPUT(P_INPUT),.V_INPUT(V_INPUT),.CLK(CLK),.RST(RST),.P_EN(P_EN),.P_BIT(P_BIT),
.TX_OUTPUT(TX_OUTPUT),.BUSY(BUSY));
    //
    initial begin
        CLK=0;
        forever #1 CLK=~CLK;
    end

    initial begin
        P_INPUT=8'b0;
        V_INPUT=0;
        RST=1; //not active
        P_EN=0;
        P_BIT=0;
        @(negedge CLK); //idle
        RST=0;
        @(negedge CLK); //idle
        RST=1; //not active
        @(negedge CLK);

        //start operation
        P_INPUT=8'b1001_0110;
        V_INPUT=0; //still in idle

        P_EN=1;
        P_BIT=0; //even parity
        V_INPUT=1; //move to start
        @(negedge CLK);
        V_INPUT=0;
        @(negedge CLK); //transition to data state
        P_INPUT=8'b1001_0110;
        repeat(11)@(negedge CLK); //reach stop
    end
endmodule
```

```

////////////////////////////////////

V_INPUT=1;
P_INPUT=8'b1111_0000;
P_EN=1;
P_BIT=1;           //odd parity
@(negedge CLK);    //start
V_INPUT=0;
@(negedge CLK);    //data
repeat(11)@(negedge CLK); //reach stop

////////////////////////////////////

V_INPUT=1;
P_INPUT=8'b0000_1111;
P_EN=0;           //skip parity
@(negedge CLK);   //start
V_INPUT=0;
repeat(10)@(negedge CLK); //reach stop

////////////////////////////////////

//sending two frames after each other (Back to Back)
P_INPUT = 8'b1010_1010;
P_EN    = 1'b0;
V_INPUT = 1'b1;

@(negedge CLK);
V_INPUT = 1'b0;

@(negedge BUSY);
@(negedge CLK);

P_INPUT = 8'b0101_0101;
V_INPUT = 1'b1;
@(negedge CLK);
V_INPUT = 1'b0;

////////////////////////////////////
// sudden reset
P_INPUT = 8'b1100_1100;
V_INPUT = 1'b1;
@(negedge CLK);
V_INPUT = 1'b0;
repeat(4) @(negedge CLK);

```

```

    RST = 0;
    @(negedge CLK);
    RST = 1;
    repeat(3) @(negedge CLK);
    $finish;
end

initial begin
    $monitor("CLK=%b | RST=%b | P_INPUT=%b | V_INPUT=%b | P_EN=%b | P_BIT=%b | TX_OUTPUT=%b | BUSY=%b ",CLK,RST,P_INPUT,V_INPUT,P_EN,P_BIT,TX_OUTPUT,BUSY);
end

endmodule

```

DO_File:

```

vlib work
vlog Main_controller_FSM.v Serializer.v mux.v paritybit_calc.v UART_TX_tb.v
vsim -voptargs=+acc work.UART_TX_tb
add wave *
run -all
#quit -sim

```

Waveform:



Questalint:

Questalint 2021.1 (D:\qformal\Projects\UART_project\lntab)

File Edit View Lint Checks Window Help

Directives

Type Filter Text...

Add Directive

Name Options Us

lint methodology soc -goal start

1 module UART_TX #(2 parameter N = 83)4)({5 input [N-1:0] P_INPUT,6 input V_INPUT,7 input CLK,8 input RST,9 input P_EN,10 input P_BIT,11 output TX_OUTPUT,12 output BUSY13);14 wire ser_done;15 wire ser_en;16 wire [1:0] mux_sel;17 wire serial_bit;18 wire parity_bit;19 wire start_bit;20 wire stop_bit;21 assign start_bit = 1'b0;22 assign stop_bit = 1'b1;23 //24 Main_controller_FSM #(.N(N)) fsm(.V_25 8, .BUSY(BUSY),26 Serializer #(.N(N)) ser(.P_INPUT(P_I_27 8

Schematic 1

CLK P_INPUT RST P_EN V_INPUT P_BIT ser done serial_data_o load ser_en fsm load mux_sel ser_en fsm load mux_sel parity_bit paritybit_calc mux mux_inout TX_OUTPUT

Flow ... D... ..

Lint Checks

Filter: Type here

Waived Fixed Pending Unimplemented Bug Verified Total: 4 Selected: 0

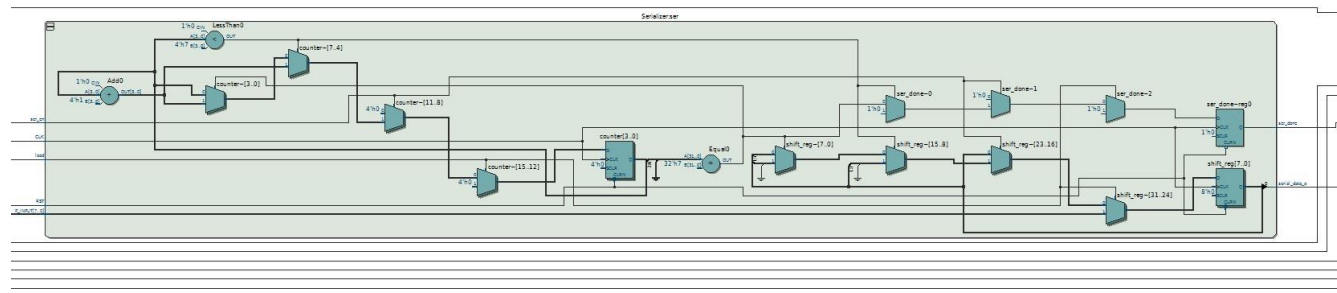
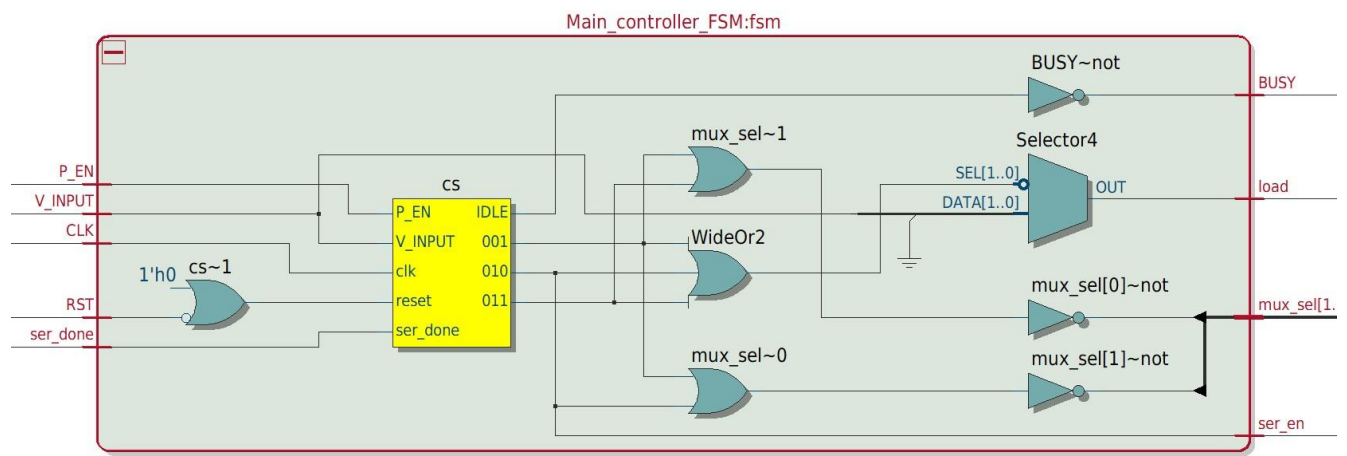
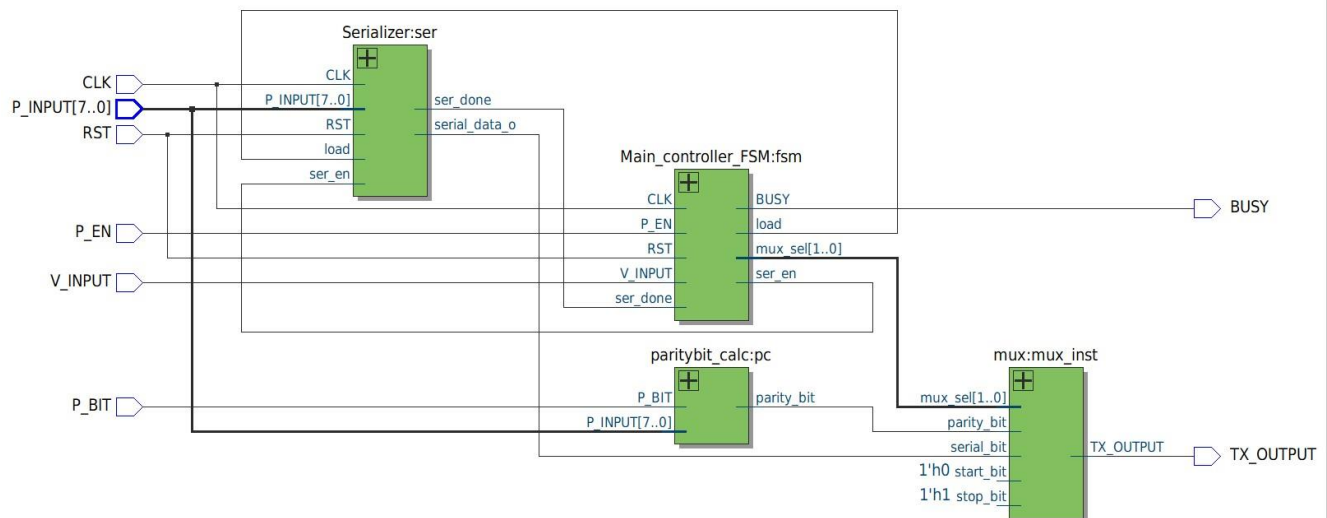
Severity	Status	Check	Alias	Message	Module	Category	State	Owner	STARC Reference
Warning	Fixed	parameter_name_duplicate		Same parameter name is used in more than one module. Parameter ...	Main_controller_F...	Nomenclature Style	resolved	ME	1.1.4.3, 3.2.2.2
Warning	Fixed	line_char_range		Line has more characters than the specified limit. Current Count 136, ...	none	Rtl Design Style	resolved	ME	3.1.4.5
Warning	Fixed	multi_ports_in_single_line		Multiple ports are declared in one line. Module Main_controller_FSM,...	Main_controller_F...	Rtl Design Style	resolved	ME	3.5.6.3
Warning	Fixed	multi_ports_in_single_line		Multiple ports are declared in one line. Module Serializer, File D:\qfor...	Serializer	Rtl Design Style	resolved	ME	3.5.6.3

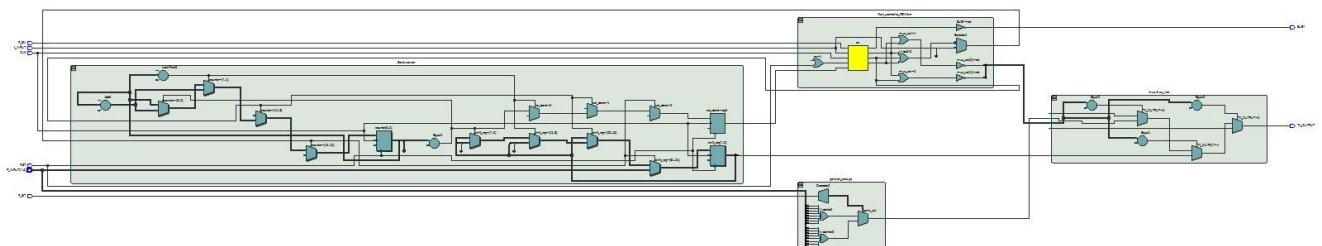
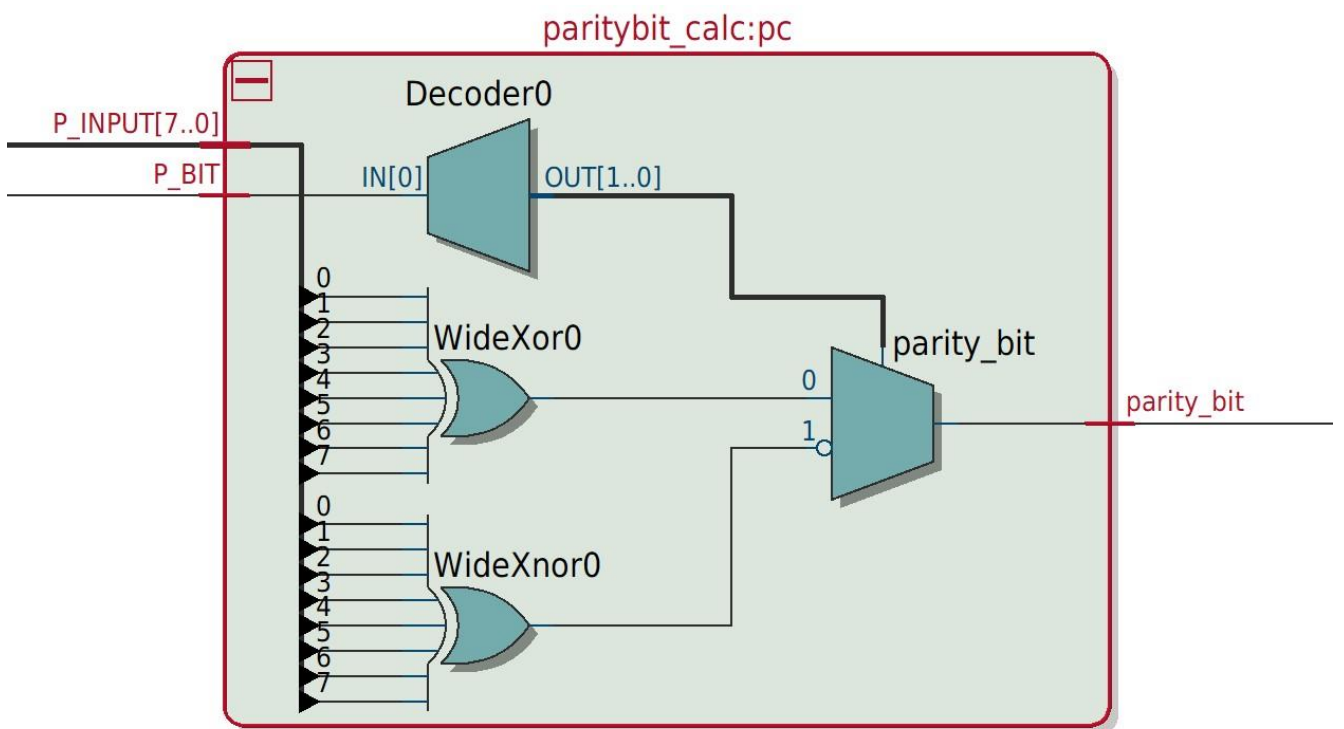
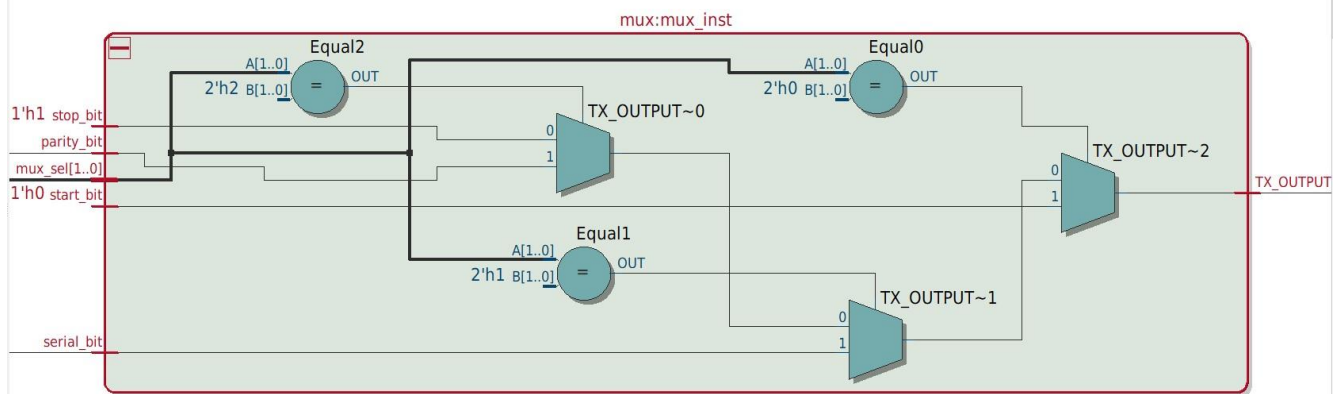
Transcript Message Viewer Lint Checks Design Metrics Design Information Status History Lint Dashboard

...mal\Projects\UART_project\UART_TX.v [UART_TX]

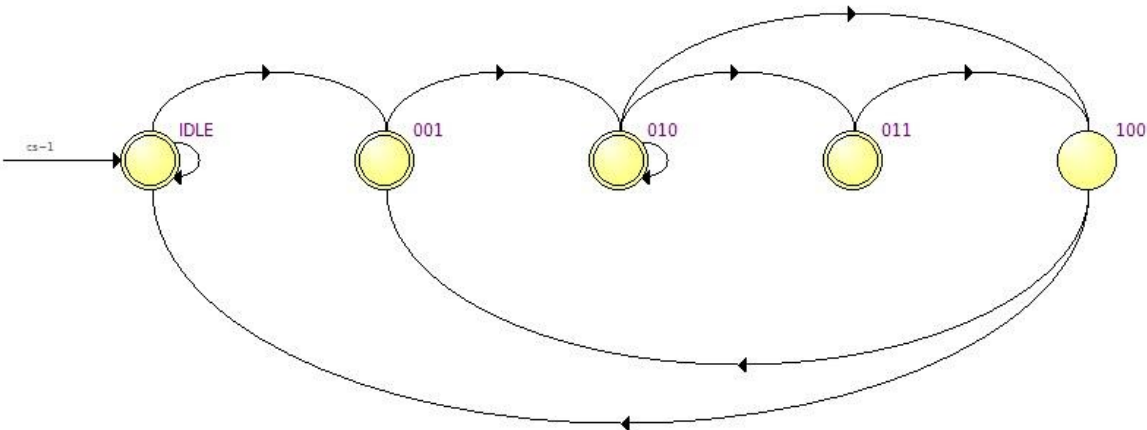
Quartus_Prime:

RTL_Schematic:





FSM_Transition:



	Name	011	010	001	IDLE	100
1	IDLE	0	0	0	0	0
2	001	0	0	1	1	0
3	010	0	1	0	1	0
4	011	1	0	0	1	0
5	100	0	0	0	1	1

	Source State	Destination State	Condition
1	001	010	
2	010	010	(!ser_done)
3	010	100	(IP_EN).(ser_done)
4	010	011	(P_EN).(ser_done)
5	011	100	
6	100	001	(V_INPUT)
7	100	IDLE	(!V_INPUT)
8	IDLE	001	(V_INPUT)
9	IDLE	IDLE	(!V_INPUT)

Timing_Report:

The screenshot displays the Quartus Prime Lite Edition interface for a project named 'UART_TX'. The main window shows the 'Timing Analyzer Summary' report, which includes a table of contents and a detailed summary of timing analysis results.

Timing Analyzer Summary

Item	Value
Quartus Prime Version	Version 25.1.0 Build 1129 10/21/2025 SC Lite Edition
Timing Analyzer	Legacy Timing Analyzer
Revision Name	UART_TX
Device Family	Cyclone V
Device Name	SC0FC7C7F23C8
Timing Models	Final
Delay Model	Slow 1100mV 85C Model
Rise/Fall Delays	Enabled

The report also includes a table of contents with the following items:

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
 - Filter
 - Assembler
 - EDA Netlist Writer
- Flow Messages
- Flow Suppressed Messages
- Timing Analyzer GUI
 - Timing Analyzer Summary
 - Advanced I/O Timing
 - SDC File List
 - Report Timing
 - Fmax Summary
 - Clocks Summary
 - Summary (Setup)
 - Summary (Hold)
 - Summary (Minimum Pulse Width)
 - Summary (Max Skew)
 - Summary (Net Delay)
 - Datasheet Report
 - Check Timing
 - Minimum Pulse Width
 - Report Skew
 - Report Max Skew
 - Report Path
 - Timing Analyzer

The console window at the bottom shows the following messages:

```
332151 Clock uncertainty is not calculated until you update the timing netlist.
332123 Deriving clock uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainties.
Found TIMING_ANALYZER_REPORT_SCRIPT_INCLUDE_DEFAULT_ANALYSIS = ON
Analyzing Slow 1100mV 85C Model
332146 Worst-case setup slack is 0.110
```

The image displays two screenshots of the Timing Analyzer software interface, showing the process of generating and viewing timing reports.

Top Screenshot:

- Menu:** The 'Reports' menu is open, showing options like 'Report Setup Summary', 'Report Hold Summary', 'Report Recovery Summary', 'Report Removal Summary', 'Report Minimum Pulse Width Summary', 'Report Max Skew Summary', and 'Report Net Delay Summary'.
- Table:** A table titled 'Summary (Hold)' is displayed with the following data:

Clock	Slack	End Point TNS
1 CLK	0.108	0.000
- Tasks:** The 'Tasks' pane shows a list of tasks including 'Read SDC File', 'Update Timing Netlist', 'Reset Design', 'Set Operating Conditions...', 'Reports', 'Slack', 'Report Setup Summary', 'Report Hold Summary', 'Report Recovery Summary', and 'Report Removal Summary'.
- Report:** The 'Report' pane shows a list of reports including 'SDC File List', 'Report Timing', 'Summary (Max Skew)', 'Summary (Net Delay)', 'Fmax Summary', 'Clocks Summary', 'Summary (Minimum Pulse Width)', 'Summary (Setup)', and 'Summary (Hold)'.
- Console:** The console shows a series of commands:


```

tcl report_max_skew -detail summary -panel_name "Summary (Max Skew)"
tcl report_net_delay -nworst 0 -panel_name "Summary (Net Delay)"
tcl report_clock_fmax_summary -panel_name "Fmax Summary"
tcl report_clocks -panel_name "Clocks Summary"
tcl create_timing_summary -mpw -panel_name "Summary (Minimum Pulse Width)"
tcl create_timing_summary -setup -panel_name "Summary (Setup)"
tcl create_timing_summary -hold -panel_name "Summary (Hold)"
      
```

Bottom Screenshot:

- Menu:** The 'Reports' menu is open, showing options like 'Report Setup Summary', 'Report Hold Summary', 'Report Recovery Summary', 'Report Removal Summary', 'Report Minimum Pulse Width Summary', 'Report Max Skew Summary', and 'Report Net Delay Summary'.
- Table:** A table titled 'Summary (Minimum Pulse Width)' is displayed with the following data:

Clock	Slack	End Point TNS
1 CLK	9.074	0.000
- Tasks:** The 'Tasks' pane shows a list of tasks including 'Read SDC File', 'Update Timing Netlist', 'Reset Design', 'Set Operating Conditions...', 'Reports', 'Slack', 'Report Setup Summary', 'Report Hold Summary', 'Report Recovery Summary', and 'Report Removal Summary'.
- Report:** The 'Report' pane shows a list of reports including 'SDC File List', 'Report Timing', 'Summary (Max Skew)', 'Summary (Net Delay)', 'Fmax Summary', 'Clocks Summary', 'Summary (Minimum Pulse Width)', 'Summary (Setup)', and 'Summary (Hold)'.
- Console:** The console shows a series of commands:


```

tcl report_net_delay -nworst 0 -panel_name "Summary (Net Delay)"
tcl report_clock_fmax_summary -panel_name "Fmax Summary"
tcl report_clocks -panel_name "Clocks Summary"
tcl create_timing_summary -mpw -panel_name "Summary (Minimum Pulse Width)"
tcl create_timing_summary -setup -panel_name "Summary (Setup)"
tcl create_timing_summary -hold -panel_name "Summary (Hold)"
tcl create_timing_summary -mpw -panel_name "Summary (Minimum Pulse Width)"
      
```

	Fmax	Restricted Fmax	Clock Name	Note
1	59.56 MHz	59.56 MHz	CLK	

Command Info		Summary of Paths						
	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	0.108	Serializer:ser counter[1]	Serial...r_done	CLK	CLK	0.000	0.030	0.138
2	0.120	Serializer:ser shift_reg[4]	Serial...reg[3]	CLK	CLK	0.000	0.018	0.138
3	0.132	Main_controll...010~DUPLICATE	Main...s.100	CLK	CLK	0.000	0.005	0.137
4	0.135	Serializer:ser ser_done	Main...s.011	CLK	CLK	0.000	0.003	0.138
5	0.135	Main_controll...IDLE~DUPLICATE	Main...s.001	CLK	CLK	0.000	0.012	0.147
6	0.136	Main_controll...IDLE~DUPLICATE	Serial...reg[6]	CLK	CLK	0.000	0.001	0.137
7	0.136	Main_controll...IDLE~DUPLICATE	Serial...reg[2]	CLK	CLK	0.000	0.001	0.137
8	0.138	Main_controll...IDLE~DUPLICATE	Serial...reg[5]	CLK	CLK	0.000	-0.001	0.137
9	0.140	Main_controll...IDLE~DUPLICATE	Serial...ter[0]	CLK	CLK	0.000	-0.001	0.139
10	0.141	Main_controll...010~DUPLICATE	Serial...ter[3]	CLK	CLK	0.000	-0.003	0.138

Path #1: Hold slack is 0.108

Path Summary	Statistics	Data Path	Waveform	Extra Fitter Information
--------------	------------	-----------	----------	--------------------------

	Property	Value
1	From Node	Serializer:ser counter[1]
2	To Node	Serializer:ser ser_done
3	Launch Clock	CLK
4	Latch Clock	CLK
5	Data Arrival Time	1.915
6	Data Required Time	1.807
7	Slack	0.108

Command Info		Summary of Paths						
	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	3.210	P_INPUT[1]	TX_OUTPUT	CLK	CLK	20.000	0.000	9.690
2	5.927	P_INPUT[2]	TX_OUTPUT	CLK	CLK	20.000	0.000	6.973
3	6.065	P_INPUT[0]	TX_OUTPUT	CLK	CLK	20.000	0.000	6.835
4	6.344	P_INPUT[6]	TX_OUTPUT	CLK	CLK	20.000	0.000	6.556
5	7.405	P_INPUT[4]	TX_OUTPUT	CLK	CLK	20.000	0.000	5.495
6	7.418	P_INPUT[5]	TX_OUTPUT	CLK	CLK	20.000	0.000	5.482
7	7.683	P_INPUT[3]	TX_OUTPUT	CLK	CLK	20.000	0.000	5.217
8	8.090	P_BIT	TX_OUTPUT	CLK	CLK	20.000	0.000	4.810
9	8.449	P_INPUT[7]	TX_OUTPUT	CLK	CLK	20.000	0.000	4.451
10	10.934	Main_controller_FSM:fsm cs.011	TX_OUTPUT	CLK	CLK	20.000	-1.910	4.056

Path #1: Setup slack is 3.210

Path Summary	Statistics	Data Path	Waveform	Extra Fitter Information
--------------	------------	-----------	----------	--------------------------

	Property	Value
1	From Node	P_INPUT[1]
2	To Node	TX_OUTPUT
3	Launch Clock	CLK
4	Latch Clock	CLK
5	Data Arrival Time	13.690
6	Data Required Time	16.900
7	Slack	3.210

Timing Analyzer - D:\DIGITAL IC\NTU\MINE\UART_project\UART_TX - UART_TX

File View Netlist Constraints Reports Script Tools Window Help

Set Operating Conditions

- Slow 1100mV BSC Model
- Slow 1100mV OC Model
- Fast 1100mV BSC Model
- Fast 1100mV OC Model

Command Info Summary of Paths

Slack	Actual Width	Required Width	Type	Clock	Clock Edge	Target
1 9.074	9.763	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[counter][0]-DUPLICATE
2 9.074	9.763	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[counter][2]
3 9.074	9.763	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[shift_reg][1]
4 9.074	9.763	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[shift_reg][3]
5 9.077	9.766	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[counter][0]
6 9.077	9.766	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[counter][1]
7 9.077	9.766	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[counter][2]-DUPLICATE
8 9.077	9.766	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[shift_reg][5]
9 9.078	9.767	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[shift_reg][0]
10 9.078	9.767	0.689	Low Pulse Width	CLK	Rise	Serializer.ser[shift_reg][2]

Path #1: slack is 9.074

Property	Data Path	Value
1 Node	Serializer.ser[counter][0]-DUPLICATE	
2 Clock	CLK	
3 Type	Low Pulse Width	
4 Actual Width		9.763
5 Required Width		0.689
6 Slack		9.074

Path #1: slack is 9.074

Property	Data Path	Value
1 Node	Serializer.ser[counter][0]-DUPLICATE	
2 Clock	CLK	
3 Type	Low Pulse Width	
4 Actual Width		9.763
5 Required Width		0.689
6 Slack		9.074

Tasks

- Read SDC File
- Update Timing Netlist
- Reset Design
- Set Operating Conditions...
- Reports
- Slack
- Report Setup Summary
- Report Hold Summary
- Report Recovery Summary
- Report Removal Summary

Report

- Summary (Setup)
- Summary (Hold)
- Summary (Minimum Pulse Width)
- Summary (Max Skew)
- Summary (Net Delay)
- Datasheet Report
- Check Timing
- Minimum Pulse Width
- Slow 1100mV BSC Model

Console

```
tbl report_max_skew -detail summary -panel_name "Summary (Max Skew)"
tbl report_net_delay -nworst 0 -panel_name "Summary (Net Delay)"
tbl report_datasheet -panel_name "Datasheet Report"
tbl check_timing -panel_name "Check Timing"
tbl report_min_pulse_width -nworst 10 -detail full_path -panel_name "Minimum Pulse Width" -multi_corner
tbl report_minimum_pulse_width: Found 10 results (0 violated). Worst case slack is 9.074
tbl
```

0% 00:00:00 Ready

Timing Analyzer - D:\DIGITAL IC\NTU\MINE\UART_project\UART_TX - UART_TX

File View Netlist Constraints Reports Script Tools Window Help

Set Operating Conditions

- Slow 1100mV BSC Model
- Slow 1100mV OC Model
- Fast 1100mV BSC Model
- Fast 1100mV OC Model

Setup Times

Data Port	Clock Port	Rise	Fall	Clock Edge	Clock Reference
1 P_EN	CLK	-0.252	-0.140	Rise	CLK
2 P_INPUT[4]	CLK	3.934	3.955	Rise	CLK
1 P_INPUT[0]	CLK	1.853	1.800	Rise	CLK
2 P_INPUT[1]	CLK	3.934	3.955	Rise	CLK
3 P_INPUT[2]	CLK	1.005	1.175	Rise	CLK
4 P_INPUT[3]	CLK	-0.174	-0.060	Rise	CLK
5 P_INPUT[4]	CLK	-0.352	-0.170	Rise	CLK
6 P_INPUT[5]	CLK	-0.346	-0.199	Rise	CLK
7 P_INPUT[6]	CLK	1.652	1.715	Rise	CLK
8 P_INPUT[7]	CLK	0.053	0.027	Rise	CLK
3 V_INPUT	CLK	0.528	0.535	Rise	CLK

Tasks

- Read SDC File
- Update Timing Netlist
- Reset Design
- Set Operating Conditions...
- Reports
- Slack
- Report Setup Summary
- Report Hold Summary
- Report Recovery Summary
- Report Removal Summary

Report

- Fmax Summary
- Clocks Summary
- Summary (Setup)
- Summary (Hold)
- Summary (Minimum Pulse Width)
- Summary (Max Skew)
- Summary (Net Delay)
- Datasheet Report
- Setup Times

Console

```
tbl create_timing_summary -setup -panel_name "Summary (Setup)"
tbl create_timing_summary -hold -panel_name "Summary (Hold)"
tbl create_timing_summary -mpw -panel_name "Summary (Minimum Pulse Width)"
tbl report_max_skew -detail summary -panel_name "Summary (Max Skew)"
tbl report_max_skew -detail summary -panel_name "Summary (Max Skew)"
tbl report_net_delay -nworst 0 -panel_name "Summary (Net Delay)"
tbl report_datasheet -panel_name "Datasheet Report"
tbl
```

0% 00:00:00 Ready

Timing Analyzer - D:\DIGITAL_IC\NTU\MINE\UART_project\UART_TX - UART_TX

File View Netlist Constraints Reports Script Tools Window Help

Set Operating Conditions

- Slow 1100mV BSC Model
- Slow 1100mV OC Model
- Fast 1100mV BSC Model
- Fast 1100mV OC Model

Tasks

- Read SDC File
- Update Timing Netlist
- Reset Design
- Set Operating Conditions...
- Reports
 - Slack
 - Report Setup Summary
 - Report Hold Summary
 - Report Recovery Summary
 - Report Removal Summary

Report

- Summary (Hold)
- Summary (Minimum Pulse Width)
- Summary (Max Skew)
- Summary (Net Delay)
- Datasheet Report
- Check Timing
- Minimum Pulse Width
- Report Skew
- Slow 1100mV BSC Model

Command Info Summary of Paths

Delay	From Node	To Node
1 9.690	P_INPUT[1]	TX_OUTPUT
2 6.973	P_INPUT[2]	TX_OUTPUT
3 6.835	P_INPUT[10]	TX_OUTPUT
4 6.556	P_INPUT[6]	TX_OUTPUT
5 5.728	P_INPUT[11]	Serializer ser[shift_reg[1]]
6 5.495	P_INPUT[4]	TX_OUTPUT
7 5.482	P_INPUT[5]	TX_OUTPUT
8 5.217	P_INPUT[3]	TX_OUTPUT
9 4.810	P_BIT	TX_OUTPUT
10 4.451	P_INPUT[7]	TX_OUTPUT

Path #1: Delay is 9.690

Path Summary	Statistics	Data Path
Property	Value	
1 From Node	P_INPUT[1]	
2 To Node	TX_OUTPUT	
3 Delay	9.690	

Path #1: Delay is 9.690

Path Summary	Statistics	Data Path
Property	Value	
1 From Node	P_INPUT[1]	
2 To Node	TX_OUTPUT	
3 Delay	9.690	

Console

```
lch check_timing -panel_name "Check Timing"
lch report_min_pulse_width -nworst 10 -detail full_path -panel_name (Minimum Pulse Width) -multi_corner
lch report_min_pulse_width: Found 10 results (0 violated). Worst case slack is 9.074
lch report_skew -npaths 1 -detail path_only -panel_name (Report Skew) -multi_corner
lch report_skew: Found 2 paths. Worst case skew is 10.488
lch
```

Timing Analyzer - D:\DIGITAL_IC\NTU\MINE\UART_project\UART_TX - UART_TX

File View Netlist Constraints Reports Script Tools Window Help

Set Operating Conditions

- Slow 1100mV BSC Model
- Slow 1100mV OC Model
- Fast 1100mV BSC Model
- Fast 1100mV OC Model

Tasks

- Read SDC File
- Update Timing Netlist
- Reset Design
- Set Operating Conditions...
- Reports
 - Slack
 - Report Setup Summary
 - Report Hold Summary
 - Report Recovery Summary
 - Report Removal Summary

Report

- Summary (Max Skew)
- Summary (Net Delay)
- Datasheet Report
- Check Timing
- Minimum Pulse Width
- Report Skew
- Report Max Skew
- Report Path
- Slow 1100mV BSC Model

Command Info Summary of Paths

Delay	From Node	To Node
1 9.690	P_INPUT[1]	TX_OUTPUT
2 6.973	P_INPUT[2]	TX_OUTPUT
3 6.835	P_INPUT[10]	TX_OUTPUT
4 6.556	P_INPUT[6]	TX_OUTPUT
5 5.728	P_INPUT[11]	Serializer ser[shift_reg[1]]
6 5.495	P_INPUT[4]	TX_OUTPUT
7 5.482	P_INPUT[5]	TX_OUTPUT
8 5.217	P_INPUT[3]	TX_OUTPUT
9 4.810	P_BIT	TX_OUTPUT
10 4.451	P_INPUT[7]	TX_OUTPUT

Path #1: Delay is 9.690

Path Summary	Statistics	Data Path
Property	Value	
1 From Node	P_INPUT[1]	
2 To Node	TX_OUTPUT	
3 Delay	9.690	

Path #1: Delay is 9.690

Path Summary	Statistics	Data Path
Property	Value	
1 From Node	P_INPUT[1]	
2 To Node	TX_OUTPUT	
3 Delay	9.690	

Console

```
lch report_max_skew -npaths 1 -detail path_only -panel_name (Report Max Skew) -multi_corner
lch report_max_skew: No paths were found
lch report_path -npaths 10 -panel_name (Report Path) -multi_corner
lch report_path: Found 10 paths. Longest delay is 9.690
lch
```

Tasks		
Task		Time
▼ Compile Design		
✓ ▶ Analysis & Synthesis		00:02:14
✓ ▶ Fitter (Place & Route)		00:03:33
✓ ▶ Assembler (Generate programming files)		00:03:01
✓ ▶ Timing Analysis		00:02:05
✓ ▶ EDA Netlist Writer		00:01:44
■ Edit Settings		
▶ Program Device (Open Programmer)		

Type	ID	Message
●	332151	Clock uncertainty is not calculated until you update the timing netlist.
●	332123	Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainties.
●		Found TIMING_ANALYZER_REPORT_SCRIPT_INCLUDE_DEFAULT_ANALYSIS = ON
●		Analyzing Slow 1100mV 85C Model
●	332146	Worst-case setup slack is 0.110
●	332146	Worst-case hold slack is 0.289
●	332140	No Recovery paths to report
●	332140	No Removal paths to report
●	332146	Worst-case minimum pulse width slack is 9.245
●		Analyzing Slow 1100mV 0C Model
●	332123	Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainties.
●	332146	Worst-case setup slack is 0.848
●	332146	Worst-case hold slack is 0.281
●	332140	No Recovery paths to report
●	332140	No Removal paths to report
●	332146	Worst-case minimum pulse width slack is 9.105
●		Analyzing Fast 1100mV 85C Model
●	332123	Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainties.
●	332146	Worst-case setup slack is 6.753
●	332146	Worst-case hold slack is 0.132
●	332140	No Recovery paths to report
●	332140	No Removal paths to report
●	332146	Worst-case minimum pulse width slack is 9.413
●		Analyzing Fast 1100mV 0C Model
●	332123	Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainties.
●	332146	Worst-case setup slack is 6.984
●	332146	Worst-case hold slack is 0.118
●	332140	No Recovery paths to report
●	332140	No Removal paths to report
●	332146	Worst-case minimum pulse width slack is 9.414
●	332101	Design is fully constrained for setup requirements
●	332101	Design is fully constrained for hold requirements
●		Quartus Prime Timing Analyzer was successful. 0 errors, 1 warning

Constraints_File:

```
# 1. Define the Primary Clock (e.g., 50 MHz -> Period = 20ns)
create_clock -name {CLK} -period 20.000 [get_ports {CLK}]

# 2. Automatically derive clock uncertainty (jitter, guardband)
derive_clock_uncertainty

# 3. Constrain Inputs (Setup/Hold timing relative to CLK)
# Assuming external inputs arrive within 2ns to 4ns after clock edge
set_input_delay -clock {CLK} -max 4.000 [get_ports {P_INPUT[*] V_INPUT P_EN P_BIT
RST}]
set_input_delay -clock {CLK} -min 2.000 [get_ports {P_INPUT[*] V_INPUT P_EN P_BIT
RST}]

# 4. Constrain Outputs (Setup/Hold requirements of receiving chip)
# Assuming external receiver requires 2ns setup time
set_output_delay -clock {CLK} -max 3.000 [get_ports {TX_OUTPUT BUSY}]
set_output_delay -clock {CLK} -min 0.500 [get_ports {TX_OUTPUT BUSY}]

# 5. Optional: Cut timing on Asynchronous Reset if handled asynchronously in design
set_false_path -from [get_ports {RST}] -to *
```